Background Shortcut Learning Mitigation for Waterbirds

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Abstract. This is our weekly report on three methods that try to stop image classifiers from relying on the background: DPE, POTER, and AutoBackSwap. We ran each method on the Waterbirds dataset and compared our numbers with the ones in the papers. DPE and AutoBackSwap came close to the published worst-group accuracy. POTER finished training for one seed, but we only have validation results so far. While looking at AutoBackSwap's mistakes, we found that three bird species seem to be labeled wrongly in the dataset, and that they cause many of its errors. We also list three ideas for improving DPE that we plan to try next week.

Keywords: Shortcut learning · Waterbirds · Reproducibility · Worst-group accuracy.

1 Introduction

1.1 Problem statement

Definition. Each image in the Waterbirds dataset [4] shows a bird, and the task is to decide whether it is a landbird or a waterbird. The difficulty is that the background is correlated with the label: most landbirds are photographed on land and most waterbirds on water. A model can therefore get good accuracy just by looking at the background, without looking at the bird.

Why it is hard. A normally trained model (ERM, which simply minimizes the average training loss) tends to learn this shortcut. It then fails on the rare images where the background is misleading, such as a landbird on water. In practice we also do not know which group an image belongs to during training, which makes the problem harder.

Research scope This week we re-ran three existing methods on Waterbirds, compared our results with the papers, and looked at some of the mistakes they make.

1.2 Literature review

The three methods we study take different approaches. POTER [2] gives more weight to training images that look like the hard cases in a small validation set. DPE [1] trains many small classifiers on top of a frozen model and averages them. AutoBackSwap [3] cuts the bird out of each image and pastes it onto a different background, so the model cannot depend on the background.

1.3 Contributions

- We ran DPE, POTER, AutoBackSwap, and an ERM baseline, and compared the results with the papers.
- We checked the individual test predictions of AutoBackSwap and ERM, and found a possible labeling problem in the dataset and a small bug in the released code.
- We wrote down three ideas for improving DPE, to be tested next week.

2 Related Work

ERM works well on average but is dominated by the majority groups. GroupDRO [4] fixes this by directly improving the worst group, but it needs group labels for every training image. The three methods in this report try to avoid that requirement.

3 Approach

All three methods use a ResNet-50 pretrained on ImageNet.

- **DPE** [1]: first train the model with ERM and freeze it. Then train 15 small classifiers (we call them heads) on balanced subsets of the data, and average their predictions.
- **POTER** [2]: compare the training images with a small validation set whose groups are known, and give higher sampling weight to the images that are similar to the hard groups.
- **AutoBackSwap** [3]: use a detector to find the bird, remove it and fill in the background, then paste the bird onto random backgrounds while training.

4 Experiments

4.1 Dataset and Metrics

We use the official Waterbirds splits [4]: 4,795 training, 1,199 validation, and 5,794 test images. The images fall into four groups: G_1 landbird on land (2,255 test images), G_2 landbird on water (2,255), G_3 waterbird on land (642), and G_4 waterbird on water (642). We report overall accuracy (Acc) and worst-group accuracy (WGA), which is the lowest accuracy among the four groups. The best checkpoint is always chosen on the validation set, never on the test set.

Table 1. Results on Waterbirds (%). Acc denotes overall accuracy and WGA denotes worst-group accuracy. DPE, ERM, and AutoBackSwap are evaluated on the held-out test split. POTER currently reflects validation metrics (epoch 258) for seed 1; held-out test evaluation is pending artifact recovery.

Method and setting	Seeds	Our result		Paper (Test)	
		Acc	WGA	Acc	WGA
DPE, standard	1	91.40	89.00	92.5±0.2	91.0±0.5
DPE, enhanced ERM*	1	93.30	92.70	96.0±0.1	94.1±0.2
POTER (val. checkpoint)	1	91.60†	90.10†	—	90.9±0.3
ERM, no minority	3	68.68±1.00	33.38±1.51	68.4±1.9	33.4±4.4
ERM, with minority	3	93.67±0.23	78.45±2.22	93.6±0.4	74.9±3.3
AutoBackSwap, no minority	3	94.49±0.40	92.33±0.16	95.1±0.3	92.2±0.4
AutoBackSwap, with minority	1	95.44	92.37	95.0±0.5	93.0±0.9

†Validation metrics at epoch 258 (held-out test evaluation pending recovery of test split logs).

4.2 Settings

DPE was run once (seed 0) in two versions: the standard one, and the enhanced one that uses a stronger ERM model (ERM*). POTER was trained once (seed 1) for 300 epochs. AutoBackSwap and ERM were run on a Tesla T4 GPU: seeds 0–2 without minority-group training images, plus one AutoBackSwap run and three ERM runs with them.

4.3 Results and Discussion

Comparison with the papers Table 1 shows our results. DPE reached 89.00% WGA in the standard setting (paper: 91.0±0.5%) and 92.70% in the enhanced setting (paper: 94.1±0.2%), so we are about 2 points below the paper in both. For ERM and AutoBackSwap, our WGA is within two standard deviations of the paper in all four settings. Without minority-group images, AutoBackSwap raises WGA from 33.38% (ERM) to 92.33% (paper: 92.2%). POTER finished training, and its best validation WGA is 90.10% at epoch 258. This is a validation number, so we cannot compare it with the paper's test WGA of 90.9% yet. We still need to recover the test results.

Failure Case Analysis (AutoBackSwap) Recall the four groups: G_1 landbird on land, G_2 landbird on water, G_3 waterbird on land, and G_4 waterbird on water. We looked at the individual test predictions of ERM and AutoBackSwap (seed 0, no minority-group images). ERM makes 1,752 mistakes, mostly on G_2 and G_3 where the background is misleading (accuracy 34.90% and 60.13%). AutoBackSwap makes only 346 mistakes: 75 in G_1, 176 in G_2, 49 in G_3, and 46 in G_4.

We noticed three more things:

- The released code prints the accuracies of G_2 and G_3 with their names swapped in `test_accuracies.txt`. After recomputing from `metadata.csv`, the real worst group is G_2 (92.20%).
- 202 test images are misclassified in all three seeds, and 109 of them are in G_2. So the errors are not random.
- Three species (`102.Western_Wood_Pewee`, `021.Eastern_Towhee`, `088.Western_Meadowlark`; 90 test images) look like landbirds, but `metadata.csv` labels them as waterbirds. AutoBackSwap calls 64 of these 90 images landbirds, which gives 64 of its 346 mistakes and 37 of the 46 mistakes in G_4. These may be label errors in the dataset rather than model errors, but we have not verified this beyond looking at the species names.

Ablation Study (DPE) In the standard setting, the full DPE gets 89.00% WGA. Using random subsets without the covariance term gives 88.20%, and using one fixed subset gives 88.80%. So the full method is better by 0.2–0.8 points, which is a small gap from a single seed. In the enhanced setting all three variants give 92.70–92.80%, so the differences are negligible.

Observations and Hypotheses on DPE We have not yet audited per-image failure cases for DPE. The observations below come from studying the method, and the hypotheses are open questions that we have not tested yet; we plan to evaluate them next week.

Observation. Each of the 15 DPE heads is a binary prototype classifier that holds exactly two prototypes, one per class. For an input feature, a head computes class probabilities from the distances to its own two prototypes only, and the final prediction is the uniform average of the 15 head probabilities. This design has three consequences: (i) prototypes are only compared within their original pair, (ii) every head contributes equally regardless of what it has learned, and (iii) each head's decision uses only 2 of the 30 available prototypes.

H1: Shuffling prototype pairs across heads. If the landbird prototype of one head is paired with the waterbird prototype of another head, does the resulting set of pairs introduce additional diversity into the ensemble and thereby improve WGA? Because the heads are trained on different balanced subsets, re-pairing yields new decision boundaries without additional training. We will compare random and exhaustive re-pairings on the trained heads, selecting the configuration by validation WGA.

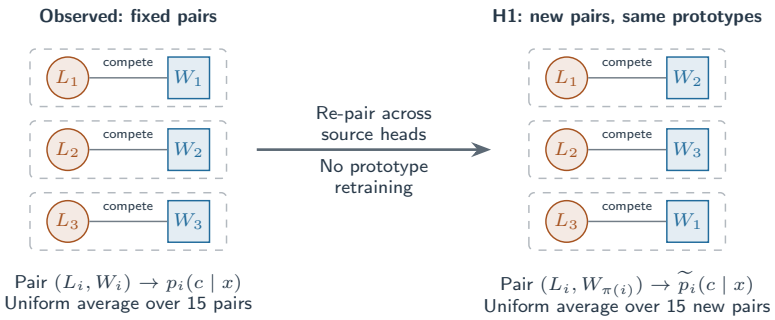


Fig. 1. H1: change which prototypes compete. Original DPE compares L_i only with W_i. An example re-pairing compares L_i with $W_{\pi(i)}$; each new pair still produces a two-class probability, and the ensemble still averages the pair probabilities. Circles denote landbird prototypes and squares waterbird prototypes; subscripts identify the source head. Only three of 15 pairs are shown. The prototypes stay fixed; whether the new comparisons improve WGA is untested.

H2: Reweighting the heads. Uniform 1/15 averaging assumes all heads are equally reliable. If some heads have learned background features and are therefore noisy, can reweighting the heads improve accuracy? We will estimate per-head weights on the validation set (e.g., from per-head worst-group accuracy) and check whether heads that receive low weights are those with low accuracy on the background-conflicting groups G_2 and G_3.

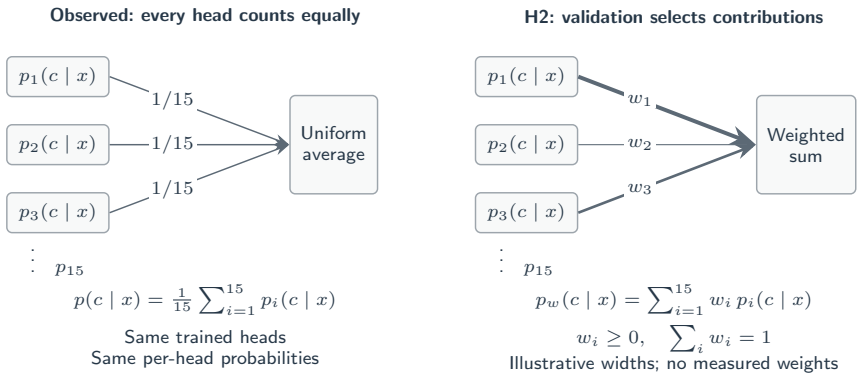


Fig. 2. H2: change how much each head contributes. The same head probabilities $p_i(c \mid x)$ enter either a uniform average or a weighted sum. Arrow widths on the right illustrate unequal contributions, not measured reliability. Nonnegative weights that sum to one would be chosen using validation data, including performance on background-conflicting groups G_2 and G_3, then fixed for test evaluation. No head has yet been identified as background-dependent.

H3: Multiple prototypes per class within each head. Can each of the 15 heads make a better prediction by using several prototypes per class instead of one? Multiple prototypes may represent different appearances within a class. We will first reuse the existing 30 trained prototypes: head i receives a class-balanced subset S_i, computes a softmax over negative-distance scores within that subset, and sums the prototype probabilities by class to obtain $p_i(L \mid x)$ and $p_i(W \mid x)$. Different heads retain different subsets. We will first average these 15 probability vectors uniformly, then evaluate the H2 weighting rule as a separate combination. Learning additional prototypes within each head would require retraining and is a possible later extension.

The original DPE is the two-prototype case, $S_i = \{L_i, W_i\}$. Global pooling is another special case: if every head uses all 30 prototypes with identical scoring and aggregation, every head produces the same probabilities, so uniform averaging or normalized H2 weighting returns the same global prediction. We will compare the original pairs, distinct larger subsets, and global pooling, selecting subset size, membership, and any score calibration using validation data only. Figure 3 illustrates these cases. All remain untested.

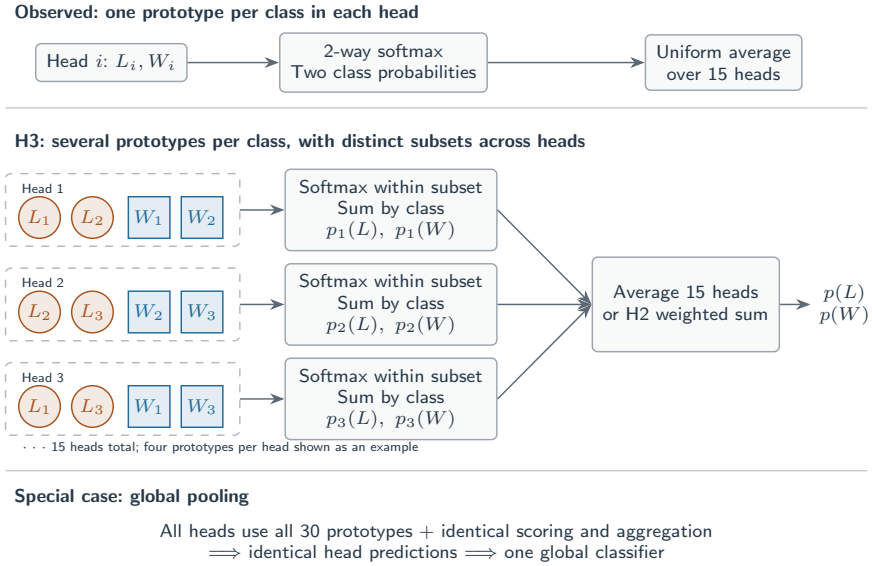


Fig. 3. H3: give each head more prototypes while retaining the ensemble. The middle panel shows two prototypes per class in three illustrative heads; all 15 heads still output two class probabilities before averaging or H2 weighting. Circles are landbird prototypes, squares are waterbird prototypes, and subscripts identify their source heads in the existing pool. Each softmax is computed within the receiving head's subset, then probability mass is summed by class. The bottom panel shows global pooling as the special case of identical full-pool heads. No accuracy gain is assumed.

5 Conclusion

DPE and AutoBackSwap gave results close to their papers, and POTER still needs its test evaluation. Looking at AutoBackSwap's mistakes, we found that three species may be mislabeled in the dataset, and that they explain about 80% of its G_4 errors. Next week we will finish POTER's test evaluation, look at DPE's failure cases, and try the three ideas above.

References

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